

Ariadna Uxue Palomino Ylla

Physics PhD Candidate | AI/ML Intern | Computer Vision | Data Science | Scientific Computing

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Profile

Physics PhD candidate and research programmer with hands-on AI/ML project experience across computer vision, biomedical statistical modeling, and scientific computing. I build reproducible Python/R workflows using PyTorch, TensorFlow/Keras, scikit-learn, statsmodels, and Git. I bring quantitative research training, clean implementation habits, and careful model evaluation to AI/ML, data science, biomedical AI, AI-agent, and physics-informed ML internships.

Technical Skills

Programming	Python, R, C, Wolfram Language/Mathematica, SPARQL, Git, basic web development (HTML, CSS, JavaScript)
Python ML/Data	PyTorch, TensorFlow/Keras, scikit-learn, pandas, NumPy, SciPy, Matplotlib, statsmodels, Ultralytics/YOLO
ML & Statistics	classification, transfer learning, CNNs/ResNet50, object detection, cross-validation, Linear Mixed Models, FDR control, model evaluation
Pipelines	preprocessing, categorical encoding, sample alignment, structured-data extraction, regex filtering, reproducible and version-controlled analysis
Scientific Computing	differential equations, numerical simulation, perturbation methods, dynamical systems, parameter studies, signal/observable analysis
Workflow	Git/GitHub, Linux, VS Code, LaTeX, Jupyter-style analysis, debugging, research documentation
Languages	English (TOEFL 110), Spanish (native), Japanese (elementary), Italian (conversational)

Selected AI/ML Projects

Computer-Vision Defect Detection Pipeline

2025–2026

Applied ML / deep learning / transfer learning / object detection

- Built an image-classification pipeline for screw defect detection with classical ML baselines, CNN transfer learning, and repeatable model-evaluation workflows.
- Fine-tuned ResNet50-style models and evaluated performance with accuracy, precision, recall, and F1 diagnostics; analyzed false positives and validation-set distribution shift.
- Ran YOLO/Ultralytics rotated-object-detection experiments to localize screws and support orientation-aware augmentation, cropping, and patch-based inference.

NeuroDE-LMM: Differential Gene Expression via Linear Mixed Models

2025–Present

Biomedical data analysis / statistical ML / reproducible Python-R pipeline

- Built a reproducible Python/R pipeline for high-dimensional differential gene-expression analysis using Linear Mixed Models.
- Implemented per-feature models to estimate diagnostic effects while controlling for donor, sample, and brain-region variation.
- Automated preprocessing, categorical encoding, sample alignment, effect-size estimation, p-value calculation, FDR correction, diagnostic visualization, volcano plots, and structured outputs.

Additional Technical Projects

Structured Data Automation and Filtering Workflows

2022–Present

Research programming / data extraction / maintainable code workflows

- Designed data-automation workflows using Wolfram Language, SPARQL, regular expressions, and Git for structured research datasets.
- Built rule-based filtering and entity-matching workflows to improve consistency across data-cleaning and information-extraction tasks.
- Maintained version-controlled research codebases with attention to reproducibility, debugging, and traceable data-processing logic.

Scientific Computing Project

Physics-Based Modeling of Accreting Black-Hole Systems

2022–2025

Scientific ML-adjacent modeling / numerical simulation / signal analysis

- Constructed perturbative background models for black holes surrounded by radial perfect-fluid inflow and evaluated several equation-of-state regimes.
- Solved governing equations for fluid behavior and test-particle geodesics; connected parameters to stability, trajectories, and observable signatures.
- Analyzed orbit evolution, apsis shift, and redshift modulation as scientific signals of environmental effects around compact objects.

Research and Programming Experience

Wolfram

Research Programmer

Remote

Jan 2022–Present

- Build data-automation tools using Wolfram Language, SPARQL, regular expressions, and Git for structured research workflows.
- Develop maintainable filtering, extraction, and validation logic for heterogeneous data sources, with emphasis on reproducibility and debugging.

Nagoya University

Doctoral Researcher, Physics

Nagoya, Japan

Oct 2024–Present

- Develop analytical and computational models for relativistic systems using differential equations, perturbation methods, and numerical parameter studies.
- Translate mathematical assumptions into testable predictions and diagnostic visualizations for stability, trajectories, and observable signals.

Pontifical Catholic University of Peru

Computer Laboratory Assistant, Programming Fundamentals

Remote

Apr 2021–Dec 2021

- Taught C programming and pseudocode to 20+ students; supported programming logic, debugging, and algorithmic problem solving.

Education

Nagoya University

PhD, Physics – Quantum Gravity Laboratory Expected Sep 2027 Dissertation: *Ringdown waves from hairy black holes*. Advisor: Professor Yoo Chulmoon. MEXT Scholarship.

Nagoya, Japan

Nagoya University

MS, Physics – Quantum Gravity Laboratory Sep 2024 GPA: 4.05. Dissertation: *Test particle motion around a black hole dressed with a spherically symmetric stationary fluid*. MEXT Scholarship.

Nagoya, Japan

Pontifical Catholic University of Peru

BS, Science with Mention in Physics Mar 2021 GPA: 4.0. Thesis: *Leptonic Modeling of the Multiwavelength Spectra of Astrophysical Sources*.

Lima, Peru

Publications

- A. U. Palomino Ylla, Y. Koga, C.-M. Yoo. *Test particle motion around a black hole dressed with a spherically symmetric stationary fluid*. Phys. Rev. D (2025). doi:10.1103/681r-bw7q
- A. U. Palomino Ylla, K. Makimo, A. Tanaka, A. Ishibashi, Y. Chul-Moon. *Ringdown waves from hairy black holes*. Manuscript under revision at Physical Review D. arXiv:2603.15979